

ENEL301 – Fundamentals of Engineering
Economics and Management

Collaboration & Teamwork

Some discussion points from this week's ethics workshop...

- Codes of (ethical) conduct likely shaped by context... (e.g. ACM clearly focussed on computing and was specific, IEEE was broad/less specific)
- What are the implications for breaching a code of (ethical) conduct?
 - If you are a member of a professional society, there is usually a process. If a breach is found, remedies can include censures, fines, suspension or expulsion from society
 - Other legal options exist...

Some discussion points from this week's ethics workshop...

NZ news | Crime

Unqualified engineer who lied on CV guilty of 38 fraud charges, fined for ethics breach

Ryan Anderson

May 13, 2022 - 11:08am

 Copy Link

- Was a student member of Engineering New Zealand
- Claimed to be an engineer , but wasn't
- Claimed to be a CPEng, but wasn't
- The fine was \$5,000 for breaching the Code of Ethical Conduct. Plus a reprimand.
- Here's the interesting bit.... this person can still claim that they are an engineer!
- 'Engineer' remains an unprotected title in NZ

What we are going to cover and when...

Week	Lectures Tuesday 10-11 & Thursday 9-10	Tutorial / Workshop Weeks 1, 4-6 Wednesday 3-4 or 4-5. Weeks 2&3 Wednesday 3-5
1	Introduction to Engineering Practice & Ethics	Ethics
	Collaboration & Teamwork	
2	Māori Perspectives in Engineering	Bicultural Competence and Confidence (timetable and attend one of B01-07 OR B08-14)
	Bias, Diversity & Ethics	
3	Intellectual Property (1/2)	
	Intellectual Property (2/2)	
4	Law in NZ (1/2)	IP Protection
	Law in NZ (2/2)	
5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life cycle assessment (1/2)	

Thoughts on James Trevelyan's podcast?



The background....

1. Engineers cannot achieve much without a lot of help from other people.
Collaboration drives performance and value.
 - You can't get anything done without collaborating with other people

Example:

As a graduate engineer, you will be involved in understanding, interpreting and negotiating technical requirements (e.g. for a design)

Without doing this, you won't be able to identify specifications, or even conceive a design!

What you might think...

- Engineers work in technical roles for the first few years and don't need to develop communication skills any further. Later, if an engineer goes into management, better communication skills will be needed.

What you might think...

- My discipline is different and I'm an exception. For example, my work will be more about programming or hands-on work.

What is known...

- Engineers can only achieve results with the help of others. Engineers need to work with and influence other people from the very start of their careers. The ability to communicate dictates the ability to collaborate.

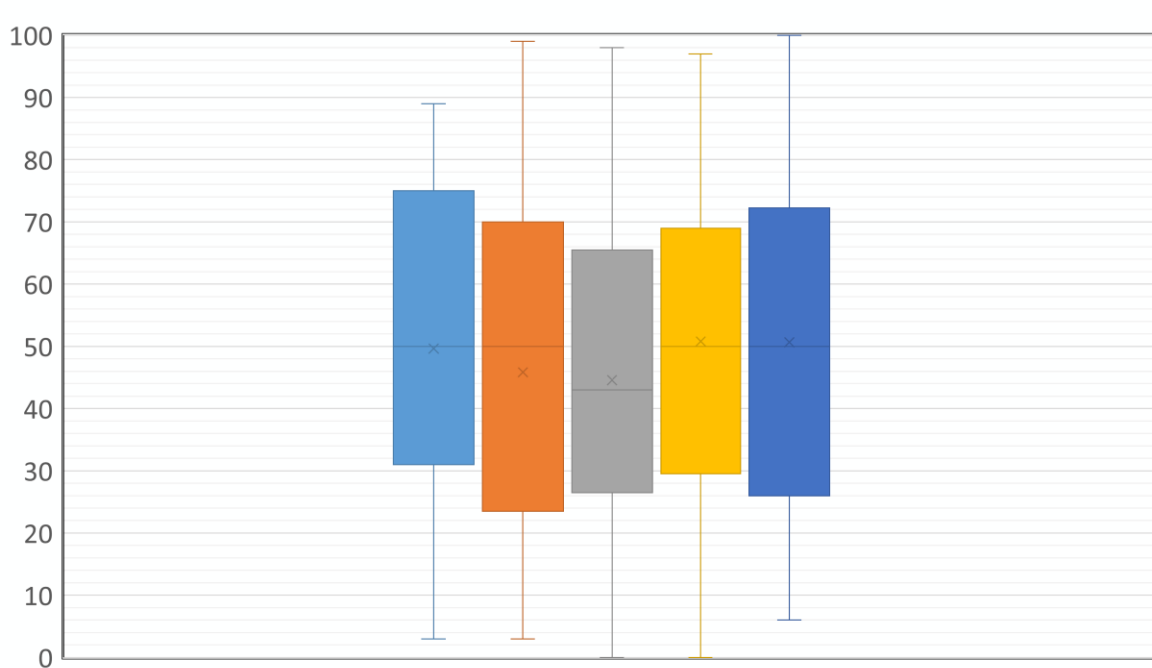
What is known...

- It doesn't matter what your discipline, you will be collaborating to get things done, including with people outside of your discipline and engineering.

Collaboration

Percentage of time spent collaborating

■ Mechatronics, n = 31 ■ Computer, n = 25 ■ Software, n = 133 ■ Electrical, n = 81 ■ Electronic, n = 54



Limitations:

- Self-reported
- Might be a real, might be an outlier
- Doesn't show effect of other factors

(Engineers also do lots of intense, focussed, individual work)

Collaboration

Working together on the same thing to purposefully exploit differences in the way we think and act through technical and social interactions*

*Adapted from Trevelyan, J. (2014), *The Making of an Expert Engineer*, 'Collaboration in engineering', pp. 233

Collaboration

Working together on the same thing

Working together on the same thing to purposefully exploit differences in the way we think and act through technical and social interactions

- Establishing process
 - Planning, establishing team charters, organising work, communication, coordination of work, reward systems etc.
- Clearly identifying tasks
 - Project planning. Who is doing what, how, when, why, to what standard etc. etc.
 - More on this later in ENEL301

Collaboration

Purposefully exploit differences

Working together on the same thing to purposefully exploit differences in the way we think and act through technical and social interactions

Collaboration

Purposefully exploit differences

Recognise and
positively exploit
difference

Collaboration Technical and Social Interactions

Working together on the same thing to purposefully exploit differences in the way we think and act through technical and social interactions

- Knowledge development is a result of your interpretation of information you receive. Much of the information you receive is through social interactions; your own experiences → tacit knowledge
- Learn how to build trust and rapport as a way of developing relationships
- Record information: listen, take really good notes, take photos / videos (if you are allowed)
- Interact: listen, eye contact, appropriate posture

Collaboration

Technical and Social Interactions

What you might think...

- It is necessary to respond to any electronic messages and e-mails immediately.

What you might think...

- E-mail and text messages are quick and convenient and you don't have to look a person in the eye when you need to convey bad news.

What is known...

- A response to a written message requires time to write and can easily be misunderstood. In fact, it is almost certain to be misunderstood. Expert engineers know how and when to respond effectively.

What is known...

- When there is any conflict or emotional arousal, there is a much better chance that any message will be interpreted in a way that further inflames the feelings of conflict.

Visit someone in person or pick up the phone;
e-mail and text messages seldom resolve conflicts

Collaboration

Technical and Social Interactions

Meetings can be the most frequent collaboration activity

Reasons to have a meeting:

- Build relationships
- Share information
- Make a decision
- Solve a problem



Collaboration

When collaboration doesn't work...

Where are some behaviours that inhibit collaboration?

- The tyranny of distance
- Inability to compromise or find common ground
- When there is a lack of respect
- Passivity / lack of input. Some don't contribute, or sit back and wait
- Wanting to be the person who solves it for the group
- Not being able to communicate technical things to non-experts
- Shutting people down / neutralising views
- Rigid mindsets
- Focussing on the small details, rather than the 'bigger picture'
- Real or perceived imbalances of members of the group
- When the money runs out
- Giving people things that they aren't good at
- Communication may be misinterpreted

Collaboration

A process perspective

Input

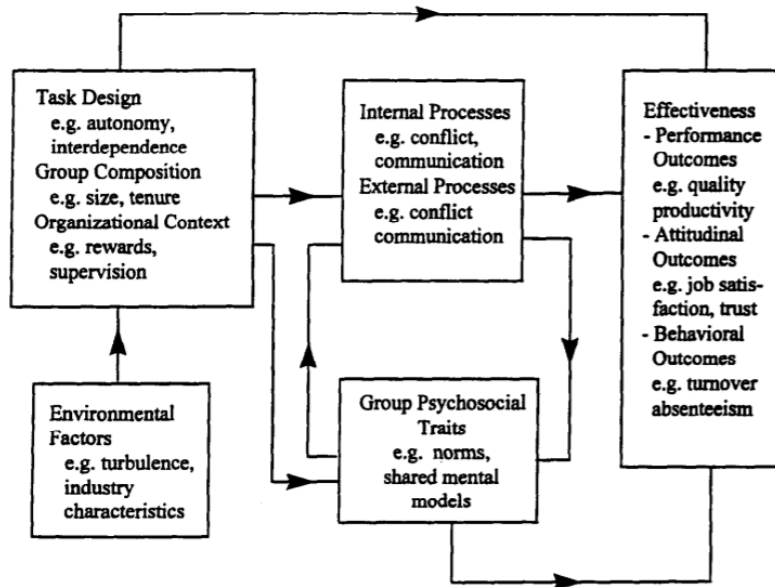


Process



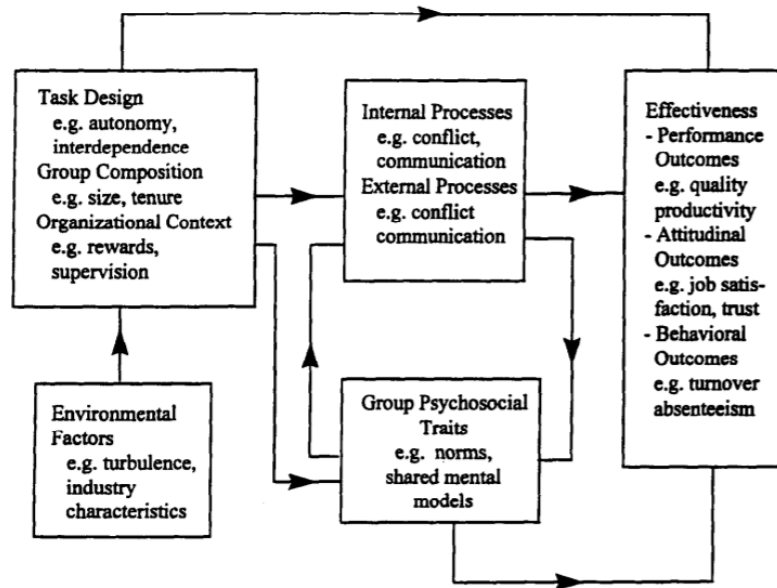
Output

- What aspects can you control?
- What aspects can you influence?
- No one has 'solved' teamwork

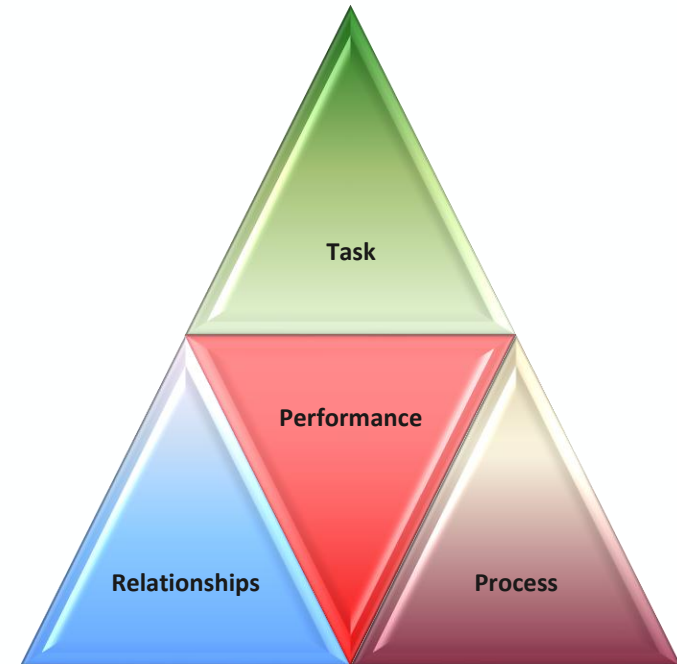


Collaboration

A process perspective



Cohen, S. G., & Bailey, D. E. (1997). What Makes Teams Work: Group Effectiveness Research from the Shop Floor to the Executive Suite. *Journal of Management*, 23(3), 239–290. <https://doi.org/10.1177/014920639702300303>

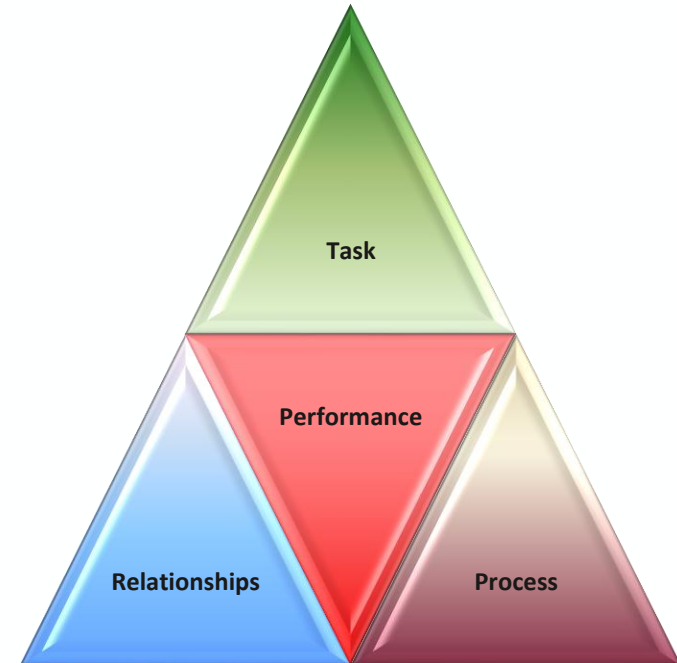


Collaboration

A process perspective

- Relationships enable engineering success
- Relationships are built on trust
- Focus on outcomes, not outputs
- Engage and build relationships with stakeholders
- Trust a path to 'social licence'

“Understanding ... culture ... is the key to getting great results in engineering” – James Trevelyan



Collaboration

Your window of opportunity

- Collaboration is typically the #1 skill that employers look for
- You've about 15 months left to develop in a collaborative environment
- Pickup the phone and speak with someone; even better, go and see them if you can!
- Collaboration in ENEL301 sustainability assignment
 - You can help shape the criteria on what makes for good collaboration
 - Two points of peer feedback. The second point can be used to adjust your assignment grade

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**Allocate yourself into Tutorial A and
Tutorial B, otherwise I'll do this for you
(deadline: Monday 20th July 9 am)**

Resources

Recommended reading:

- Trevelyan, J. P. (2014). Collaboration in engineering. In *The Making of an Expert Engineer* (pp. 189–247). CRC Press. [Ebook available from the library](#)

Recommended podcast:

- [In Conversation with Dr. Mike Cook: building confidence in a changing landscape](#)
 - Structural engineering focus, but it remains highly relevant for all disciplines

Links available on LEARN